

SciOdyssey.

Autonomous ML Research Agent

Preserve the **research world**. Reset the researcher.

01 / METHOD

Different lifetimes. Shared evidence.

Long-lived reasoning can anchor the search; rebuilding context is costly. SciOdyssey retains the world while reopening judgment with a fresh Scientist.

Research environment

Task contract + research preferences

Data + evaluator + baseline · budget + permissions

↓

Research world

PERSISTENT / ON DISK

EVIDENCE

Knowledge + experiment ledger

Metrics, failures and original reports

CURATED

Research brief + fallible priors

Current understanding; open to revision

CODE

Retained implementation + State

Recoverable version with evidence and provenance

↓ CONTEXT RESET

Inherit experience, not prior reasoning.

Scientist

SCIENTIFIC JUDGMENT

Read the research world

Form hypotheses

Code, train and evaluate

Interpret + report evidence

Chooses the science within one trajectory.

→

META

PERSISTENCE AUTHORITY

Audit claims ↔ evidence

Scope conclusions

Curate shared memory

Preserve optional State

Maintains what survives across trajectories.

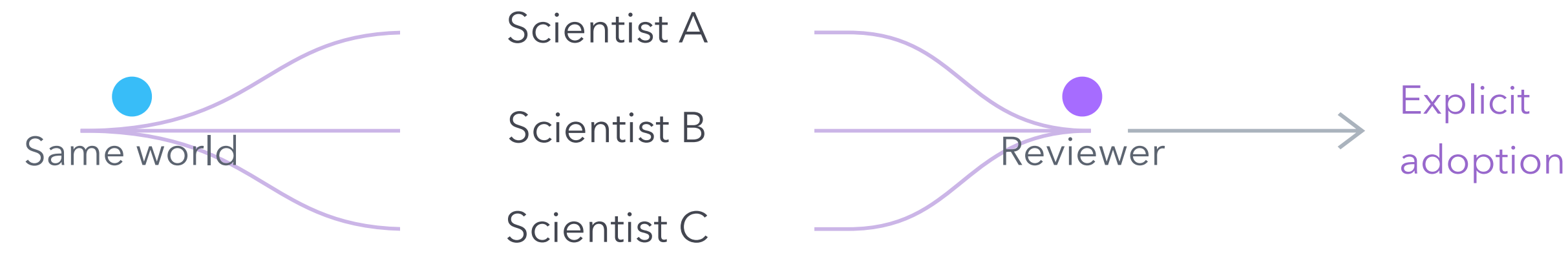
Audit → scope → preserve → fresh Scientist

Runtime / deterministic mechanics

Isolation · cancellation · runner + evaluator

Logging · artifact capture · State operations

One world. Several independent searches.



Isolated worktrees, then post-hoc review. Optional synthesis keeps one implementation parent and reference evidence; code and memory are not auto-merged.

Framework capability, not an isolated cause of benchmark gain.

03 / YOUR RESEARCH WORKFLOW

One agent. Any ML task. End to end.

task.md

PERSONAL.md

What to solve

How you want it solved

Objectives · constraints · evaluation

Preferences · research style · priorities

Define the task

→ Delegate research

→ Review evidence

github.com/zc6600/research-agent-kuairand

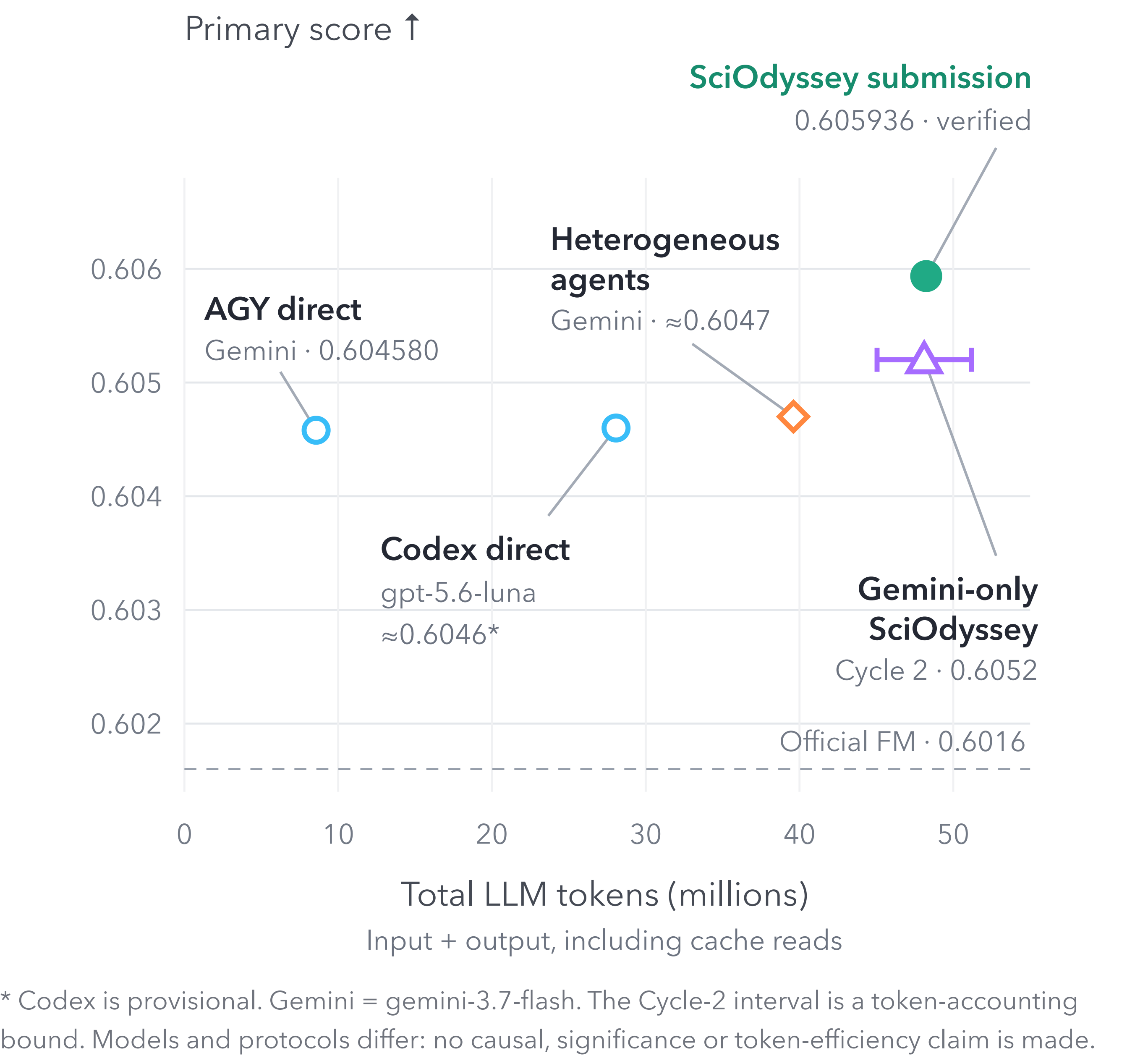
Chen Zhu · Zhou Ziyu · Shilin Xu · GE GAO · Jiran Li

Sources: slides/slides.md · docs/ARCHITECTURE.md · docs/FINAL_REPORT.md §§3-5 · Code, report, ledger and checked output in repository.

02 / EVIDENCE

Research outcomes, in context.

KuaiRand-Pure · public validation · one recorded run per condition



Verified submission · public validation

Metric	Official FM	SciOdyssey	Absolute Δ
Primary	0.6016000	0.6059363	+0.0043363
GAUC	0.6674000	0.6728421	+0.0054421
nDCG@5	0.5357000	0.5390304	+0.0033304

Primary = mean(GAUC, nDCG@5). Final recipe: 46 categorical features × eight-seed FM ensemble · NumPy / CPU.

4

autonomous cycles

13

named experiments

0

manual scientific interventions*

7 Full evaluations · 0 GPU-hours · 170,588 prediction rows passed the unchanged Starter Kit alignment checker.

* After launch, in the retained submission record.

Observed behavior, not just a score.

Recovery

After evaluation, NumPy float32 values broke serialization. The Scientist repaired the writer and retained the evidence.

Audit

Reviews found leakage and sampling mismatches. META scopes evidence; it is not a formal verifier.

Open questions. One task does not establish generality. META, State and reset effects are not isolated; summaries can still lose context and anchor future work.

Autonomous by default. Supervisable by design.

Coding agent

EXAMPLE INPUT

Use the research-agent skill for task.md.

Follow PERSONAL.md. Run autonomously.

Let me review the results and evidence.